

AI Audit Report (HW04 §9) — Features A / B / C

Faculty of Information Technology (FIT) – Ho Chi Minh City University of Science (HCMUS)
CS423 / CSC13003 – Software Testing (AI-augmented · 2026)
Assignment: HW04 – Automation Testing · Features A (FR-03), B (FR-08), C (FR-15)

1. Student Information

Field	Value
Student name	Vo Ngoc Bich Tram
Student ID	23127271
Class / Cohort	23KTPM3
Assignment ID	HW04
Assignment title	Automation Testing (Playwright)
Features in scope	A FR-03 · B FR-08 · C FR-15
Assignment date	2026-08-07 → 2026-08-10
AI tool(s) used	Cursor Agent (Grok / Composer in Cursor IDE)
AI was used	Yes

Disclosure statement: I use AI tools for the following tasks (detailed in §2).
Allowed tools (HW04 §8): Cursor Agent · Playwright · Playwright HTML reporter.

2. Audit Table

Full stage prompts / outcomes: ai-conversion-log.md , prompt_log.md .

Feature A — FR-03 Forgot Password

#	(1) Tool · Date · Prompt	(2) AI output summary	(3) Verdict	(4) Reasoning	(5) Student fix
1	Cursor · 2026-08-07 · Analyze FR-03 from README + HW04 PDF; establish contract (student ID, ≥12 cases, data-driven, 3 browsers, labeled HTML)	Actors/steps/oracles; noted SUT gaps (4-digit OTP, no confirm, no step indicator)	VALID	Matches HW04 Task 1 + README FR-03 without inventing StudentID	Kept unique API users; did not mutate test@eshop.com
2	Cursor · 2026-08-07 · Design ≥12 FR-03 cases (pos/neg/boundary/UI)	Draft TC-FORGOT-001...014 mix	INCOMPLETE	Good breadth; first draft risked asserting SUT quirks instead of spec for UI contracts	Forced 010–014 oracles to spec so defects stay visible
3	Cursor · 2026-08-07 · External JSON schema +	fr03-forgot-password.json +	VALID	Satisfies “separate json; no	Raised minCases to 12; reject

#	(1) Tool · Date · Prompt	(2) AI output summary	(3) Verdict	(4) Reasoning	(5) Student fix
	loader (no inline cases)	load-test-data.js with journey/assertion vocabulary		hardcoded case arrays"	unknown assertion types
4	Cursor · 2026-08-07 · Generate Playwright page object + data-driven spec + 3-browser matrix	Spec, ForgotPasswordPage , run-matrix.js , config	INCOMPLETE	Fragile spots: sync alert() deadlock; post-nav visibility asserts; broad "đăng nhập" locator	Human review repairs in row 5
5	Cursor · 2026-08-07 · Verify list → Chromium → matrix; repair failures that are automation bugs	Chromium then full matrix; stamped Run by: 23127271	INCOMPLETE → accepted	Product fails 010– 014 kept; automation bugs fixed (dialog Promise.all , end-state asserts, tight back-link)	Matrix: 9 pass / 5 fail × 3 browsers
6	Cursor · 2026-08-09 · Double-check HW04 PDF gaps; write AI Audit + Critique + formal bugs	Gap analysis, this audit, critique, bug-report drafts	INCOMPLETE	Audit/critique mandatory (§9– §10). GitHub Issues still student- owned (§6 / §11)	Student opens Issues + attach PNGs

Feature B — FR-08 Checkout

#	(1) Tool · Date · Prompt	(2) AI output summary	(3) Verdict	(4) Reasoning	(5) Student fix
7	Cursor · 2026-08-09 · Prep Feature B without overwriting Feature A evidence	Freeze archive + EVIDENCE- LOCK; test:matrix:fr08 ; merge-safe manifests; default adhoc report slug	VALID	Protects A HTML timestamps (Anti- cheat §11)	Run evidence:verify- fr03 before/after B work
8	Cursor · 2026-08-09 · Analyze + Design FR-08 ≥ 12 cases	Auth gate, empty cart, totals, editable amount, client- tampered total, cart clear	VALID	Matches README FR-08 + payment integrity rules	Kept 14 IDs; no padding
9	Cursor · 2026-08-09 · Model data + Generate data-driven checkout suite	fr08-checkout.json , CheckoutPage , fr08- checkout.spec.js	INCOMPLETE	First cart seed via storage alone left SPA cart empty → false fails	Seed cart through UI/API path the SPA actually reads; keep oracles
10	Cursor · 2026-08-09 · Verify 3- browser matrix; file BUG-FR08	Matrix 9 pass / 5 fail × 3; BUG-FR08-001...005	VALID (after seed fix)	Failures are product defects (auth, empty cart, editable/trusted total, cart clear)	Did not soften expected totals or auth gate

Feature C — FR-15 Product CRUD (Admin)

#	(1) Tool · Date · Prompt	(2) AI output summary	(3) Verdict	(4) Reasoning	(5) Student fix
11	Cursor · 2026-08-10 · Analyze only (no cases/code)	fr15-analysis.md : actors, CRUD, Tên/Giá/Danh mục, FR-12 linkage, ambiguities	VALID	Stage-gated; no premature Generate	Accepted Analysis; next = Design
12	Cursor · 2026-08-10 · Design ≥12 cases only	14 cases TC-PRODUCT-001...014 in fr15-design.md	VALID	Mix CRUD + validation + FR-12 auth	Review next
13	Cursor · 2026-08-10 · Review oracles / likely fails	Kept 14; forecast ~6 pass / 8 defect fails	VALID	Oracles mapped to UI/API/status; no code	Accept Review
14	Cursor · 2026-08-10 · Model data JSON only	fr15-admin-product.json (14); credentials via authMode only	VALID	No selectors/secrets in JSON	Map → Generate
15	Cursor · 2026-08-10 · Map automation	Locators, setup/cleanup, action/expect vocabulary	VALID	Spec still deferred	Generate when ready
16	Cursor · 2026-08-10 · Generate page object + data-driven spec	AdminProductPage , product-api.js , fr15-admin-product.spec.js , matrix :5174	VALID	Oracles not softened for known SUT bugs	Discovery 42 tests
17	Cursor · 2026-08-10 · Verify Chromium then full matrix	Chromium 6/8; product fails 004/006/008–010/012–014; BUG-FR15 drafted	VALID (product fails kept)	Matches Review forecast; no oracle softening	File Issues + keep matrix HTML

3. Summary of AI Accuracy

Metric	Count	%
Interactions audited (A+B+C)	17	100%
VALID as-is	12	~71%
INVALID	0	0%
INCOMPLETE (accepted after edits)	5	~29%

4. What the AI got wrong (human review — HW04 Task 1)

Issue	Feature	Why AI missed it	Fix
Sync alert() on weak password blocked Playwright click()	A	Assumed async dialogs	Promise.all(dialog, action) + noWaitAfter
Happy-path asserted step-1 controls after /login	A	Checklist asserts without lifecycle phases	End-state oracles only for full reset
Locator /đăng nhập/i matched navbar	A	Over-broad regex	Exact ^Quay lại đăng nhập\$
Temptation to set OTP length expectation to 4	A	Matching SUT hides SEC-07 defect	Keep expect 6

Issue	Feature	Why AI missed it	Fix
Cart seed via storage left SPA cart empty	B	Ignored how React cart hydrates	Seed through path SPA reads
Temptation to accept HTTP 200 on invalid product fields	C	Prefer green demos over FR-15/FR-12	Keep 400–499 / 401–403 oracles
Stage-skipping risk (Generate before Review)	C	Speed bias	Enforced Analyze→...→Verify prompts

5. Conclusion

AI was effective for stage-gated scaffolding across three features: external JSON, page objects, matrix reporting, and first-pass case lists. It was weak on browser dialog timing, SPA state seeding, locator precision, and staying loyal to **spec oracles** when the SUT is defective. For HW04, AI drafts must be followed by headed/matrix execution and an explicit “do not soften failing oracles” review.

Use AI for: stage-by-stage conversion, JSON schemas, page objects, report stamping.

Do not rely on AI alone for: final Pass/Fail interpretation, GitHub Issue authorship ethics, or claiming green matrix by changing expected results.

6. Mandatory Disclosure

I used **Cursor Agent** to analyze FR-03, FR-08, and FR-15; design/automate 14 data-driven cases per feature; configure Chromium/Firefox/WebKit HTML reports labeled `Run by: 23127271`; repair automation defects (dialogs, SPA cart seed, locators); and draft this AI Audit / Critique / gap analysis / bug reports. I reviewed outputs against the EShop README and HW04 PDF Task 1. I will not submit raw AI output without review. HTML reports and timestamps are from real local runs (Anti-AI-Cheat §11).

Appendix: `prompt_log.md` · stage log: `ai-conversion-log.md` · gaps: `hw04-pdf-gap-checklist.md`.

Signature

Field	Value
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Student ID	23127271
Class / Cohort	23KTPM3
Course	CS423 / CSC13003 – Software Testing
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Date	2026-08-10
Signature	Tram